POWER PULSE

Power Factor Control Important Formulae

Notes by Balaji Sir

1 Power triangle

1. $$P = \sqrt{3} V_L I_L \cos\phi, \quad Q = \sqrt{3} V_L I_L \sin\phi, \quad S = \sqrt{P^2 + Q^2}$$

2. $$\cos\phi = \frac{P}{S}, \qquad \tan\phi = \frac{Q}{P}$$

2 Improvement by capacitors

3. $$Q_c = P(\tan\phi_1 - \tan\phi_2)$$

kVAr of capacitor for pf change cosφ₁ → cosφ₂

4. $$C_\Delta = \frac{Q_c}{3\,\omega V_L^2}, \qquad C_Y = \frac{Q_c}{\omega V_L^2}$$

per phase (Q_c total 3-phase)

5. $$I_c = \frac{Q_c}{\sqrt{3} V_L}$$

6. $$\text{kVA reduction} = \frac{P}{\cos\phi_1} - \frac{P}{\cos\phi_2}$$

maximum-demand charges fall

7. $$\text{Loss} = 3I^2R = \frac{P^2 R}{V^2 \cos^2\phi} \Rightarrow \text{loss} \propto \frac{1}{\cos^2\phi}$$

3 Synchronous motor and combined loads

8. $$P_{total} = \sum P_i, \quad Q_{total} = \sum Q_i, \quad \cos\phi = \frac{P_{total}}{\sqrt{P_{total}^2 + Q_{total}^2}}$$

lagging loads +Q, leading loads −Q

9. $$Q_{motor} = \sqrt{3} V_L I \sin\phi \ \ (\text{leading, over-excited})$$

a synchronous motor can supply vars while giving mechanical power

10. $$\Delta V \approx \frac{PR + (Q - Q_c)X}{V}$$

capacitor also reduces voltage drop

11. Over-compensation gives a leading pf and a voltage rise